Defence presentation outline

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1. Content (1 min)
    (a) Intro, chapter 1, chapter 2, conclusion; will last 15-20 minutes.
2. Introduction
    (a) ACC lengthens growing seasons
        i. Happens through earlier SOS + delayed EOS
        ii. (Figure): conceptual figure simplified showing SOS and EOS shifts
    (b) GS are longer, but also warmer
        i. Total annual GDD increases
        ii. (Figure): conceptual figure with GDD accumulation curve
    (c) SOS + EOS shifts = more GDD, but shifts are assymetric
        i. Impacts how much GDD is accumulated in the GS
        ii. SOS shift more, but are cooler
        iii. EOS shift less, but are warmer
        iv. (Figure): conceptual figure with showing how much GDD in each season
    (d) SOS + EOS shifts = longer and warmer seasons; should increase growth
        i. Recent studies failed to find this relationship
        ii. Important because temperate forest= 40% of global net carbon sink
        iii. (Figure): TBD
    (e) Question: do trees grow more with longer and warmer seasons?
        i. Objectives: address GS/growth disconnection using two systems
        ii. Chapter 1: juvenile trees, 4 species, 4 prov, leaf phenology and tree rings
        iii. Chapter 2: mature trees, 11 species, leaf phenology and tree rings
        iv. (Figure): simplified version of thesis overview figure
3. Chapter 1
    (a) Introduction:
        i. Drought and competition could constrain growth
        ii. Internal constraints (local adaptation?)
        iii. Answer if juvenile trees grow more in a drought and competition limited environment
        iv. (Figure):
    (b) Methods: common garden set up
        i. Trees grown from seeds, 4 species, 4 provenances
        ii. Phenology for 3 years
        iii. Calendar and thermal season from leafout to budset
        iv. Growth via tree rings
        v. (Figure): map and common garden photo
    (c) Methods: stats?
        i. Bayes hierarchical model

        ii. 4 different season predictors
        iii. (Figure): model equation and conceptual figure showing predictors?
    (d) R/D: thermal > calendar
        i. Species grew more with warmer seasons than longer calendar season
        ii. Because warmer is a better predictor of growth
        iii. Thermal season weights growth days while calendar assumes a stable growth rate throughout the season
        iv. (Figure): 2 pannel mu plot for GDD and GSL
    (e) R/D: EOS > SOS;
        i. EOS temperature higher temperature than SOS
        ii. EOS delayed less, but contribution is higher because days are warmer
        iii. Suggests looking more at EOS than SOS
        iv. (Figure): 2 pannel mu plot for EOS and SOS AND conceptual figure showing GDD differences at EOS and SOS
    (f) R/D: No provenance effect;
        i. In this small climatic gradient, no evidence of local adaptation on growth
        ii. (Figure): provenance mu plot

4. Chapter 2
    (a) Introduction: life history strategies could inform tree growth responses
        i. Fast vs slower growing species
        ii. Diffuse vs ring porous species
        iii. Carbon allocation to storage = blur annual growth response to thermal and calendar seasons
    (b) Methods: citizen science leaf phenology at the Arboretum
        i. Growth via tree rings
        ii. Leaf phenology
        iii. 11 angiosperm deciduous species spanning different growth strategies and wood porousness
        iv. (Figure): Arboretum map AND core scans
    (c) Methods: stats
        i. Hierarchical model for the 4 predictors
        ii. Leafout and leaf coloring as SOS and EOS
        iii. Previous year model
        iv. (Figure): TBD
    (d) R/D: weak response across species across current year predictors
        i. Ball decreased growth: southernmost range reduction in future ACC
        ii. Tiam: grew less with earlier SOS despite being fast growing
        iii. No consistent trend across life history strategies or wood porousness
        iv. (Figure): mu plots 4 facet
    (e) R/D: 2 species responded to previous year thermal season
        i. Ball increase growth with previous yr condition
        ii. Bnig grew more only when considering previous yr condition
        iii. Reveals importance of carbon allocation to storage for mature trees; blur responses
        iv. (Figure): previous year conditions mu plot

5. Conclusion and future work
    (a) Chapter 1 and 2
        i. Chapter 1: warmer seasons for juvenile trees = good for forest regeneration and range expansion
        ii. Chapter 2: no effect of longer and warmer seasons: probably bad for forest carbon
        iii. (Figure): simplified version of thesis overview figure
    (b) Fuelinex
        i. Observational studies have limits
        ii. Fuelinex will look at carry over effect of longer seasons with relative effect of each season.
        iii. (Figure): some random photo of the experiment